

(a)

S1: Baseline

(b)

S2: Centralized BIM+DT

(c)

S3: Tokenized BIM-Twin

Disruption Probability

0.765

0.292

0.708

0.742

0.187

0.628

0.591

0.199

0.653

Disruption Severity

-0.050

0.055

-0.110

-0.019

-0.031

-0.065

-0.016

0.054

-0.120

Worker Efficiency

-0.160

-0.078

-0.124

-0.237

-0.112

-0.070

-0.173

-0.089

-0.079

SC Response Time (hrs)

-0.031

0.044

-0.047

-0.129

0.028

-0.060

-0.019

0.023

0.081

Max Concurrent Disruptions

-0.063

-0.046

-0.085

-0.030

-0.064

0.037

0.121

-0.023

-0.066

Material Volatility

0.098

0.826

0.098

0.096

0.844

0.040

-0.032

0.837

0.061

Absenteeism Rate

-0.052

-0.097

0.102

0.035

0.031

0.140

-0.058

-0.020

-0.124

Equipment Failure Prob.

0.020

0.050

-0.021

-0.090

0.037

-0.019

0.116

-0.053

0.172

Schedule
Overrun (%)Cost
Overrun (%)Idle
Rate (%)Schedule
Overrun (%)Cost
Overrun (%)Idle
Rate (%)Schedule
Overrun (%)Cost
Overrun (%)Idle
Rate (%)

Pearson r

0.8

0.6

0.4

0.2

0.0

-0.2

-0.4